

HALL A MØLLER POLARIMETER UPDATE



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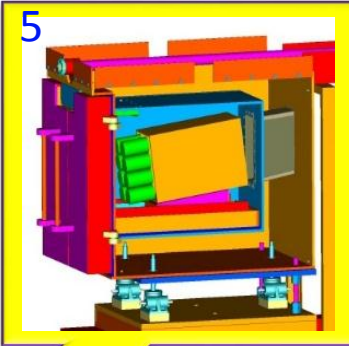
J. Gomez
Jefferson Lab

11 GeV Upgrade

6



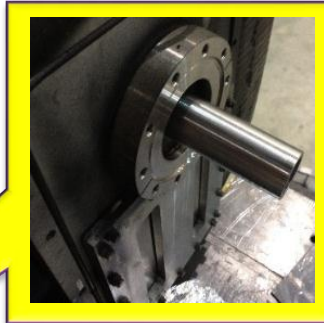
5



$10^\circ \rightarrow 7.3^\circ$



4



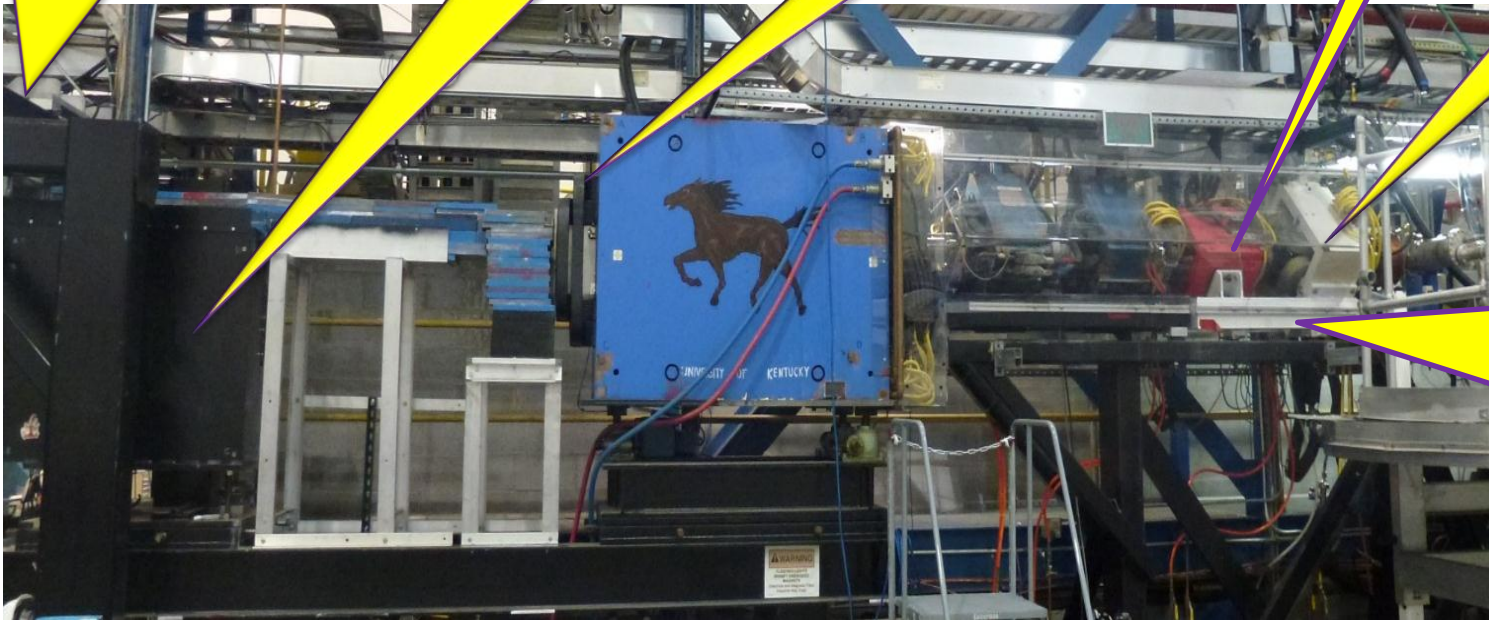
2



1



3



- 4 - Shielding pipe in dipole
- 5 - New detector shield box
- 6 - Reconfigured beamline:
v. corrector + quad + BPM

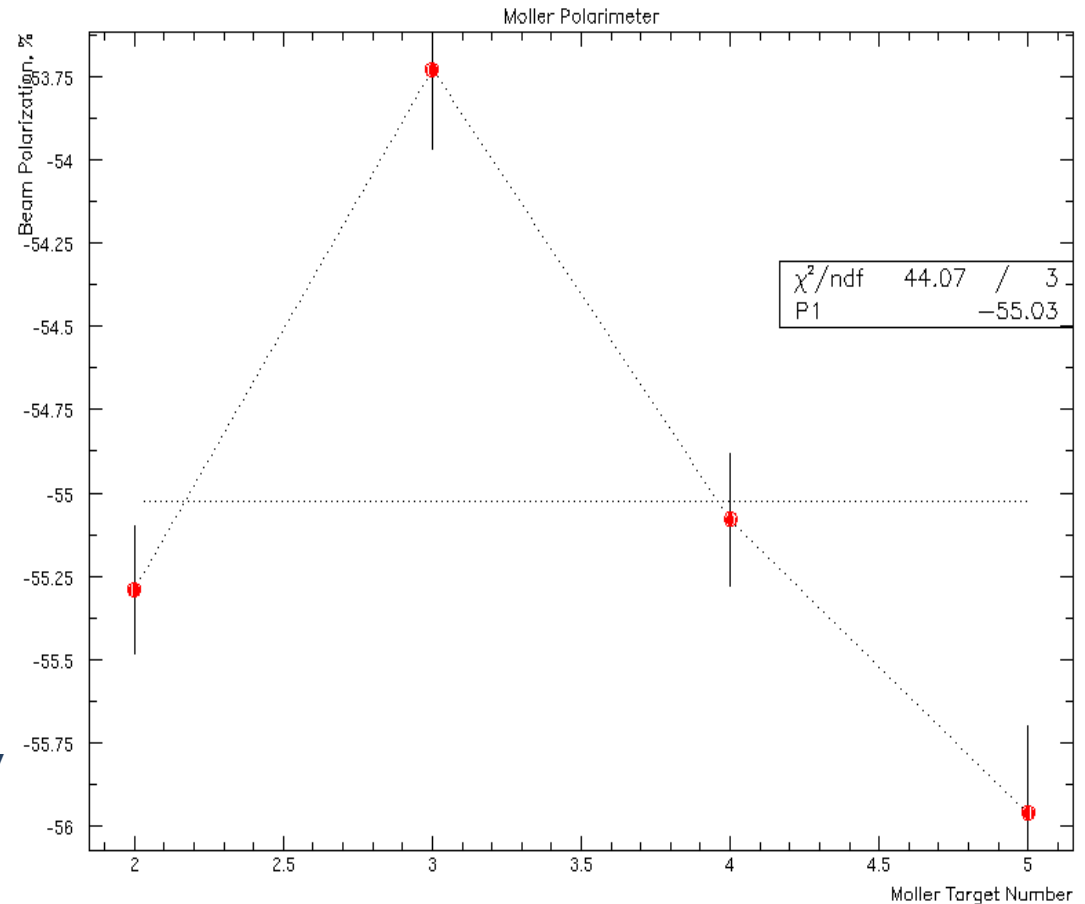
- 1 - 1st quad shifted downstream
- 2 - New 4th quad added
- 3 - New girder for quads #1 and #4

COMMISSIONING RUN (April 02 – 03, 2014)

Beam energy: 6054 - 6113 MeV.
Beam current: 0.2 - 0.5 μA .
Wien filter angle: 0° .
Helicity: 30Hz (1kHz).
Both old DAQ and FADC DAQ
were running in parallel.

Predicted polarization for
Hall A is $\sim 66\%$
with Beam energy: 6112.85 MeV
Injector polarization: $\sim 85\%$
Wien filter: 0°

Moller result 04/02–03/2014



Target Position	6	5	4	3	2	1	0
Material	park	SM	Fe	Fe	SM	SM	Al
Thickness, μm		6.8	9.3	14.3	29.4	13.0	16.5

COMMISSIONING RUN (December 08, 2014)

Moller result 12/08/2014

Beam energy: 7290 - 7294 MeV.

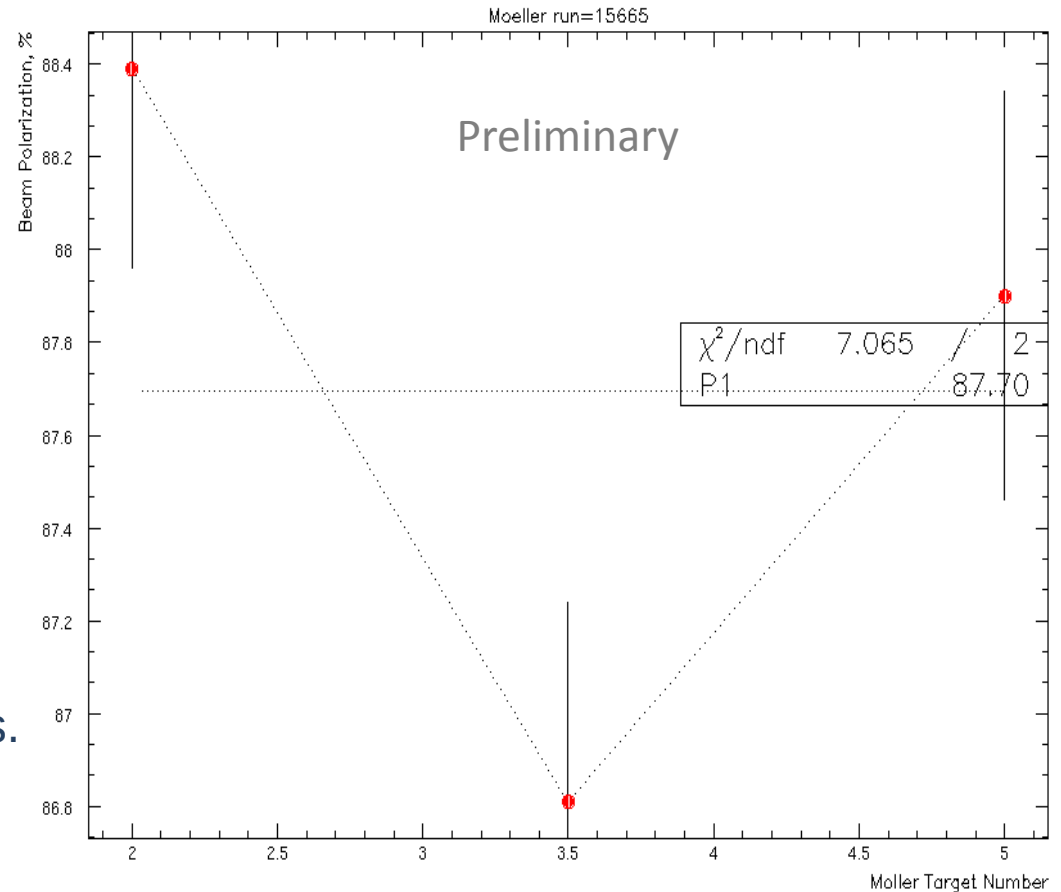
Beam current: 0.4 - 0.5 μ A.

Wien filter angle: 21.6° .

Helicity: 30 Hz.

Both old DAQ and FADC DAQ were running in parallel.

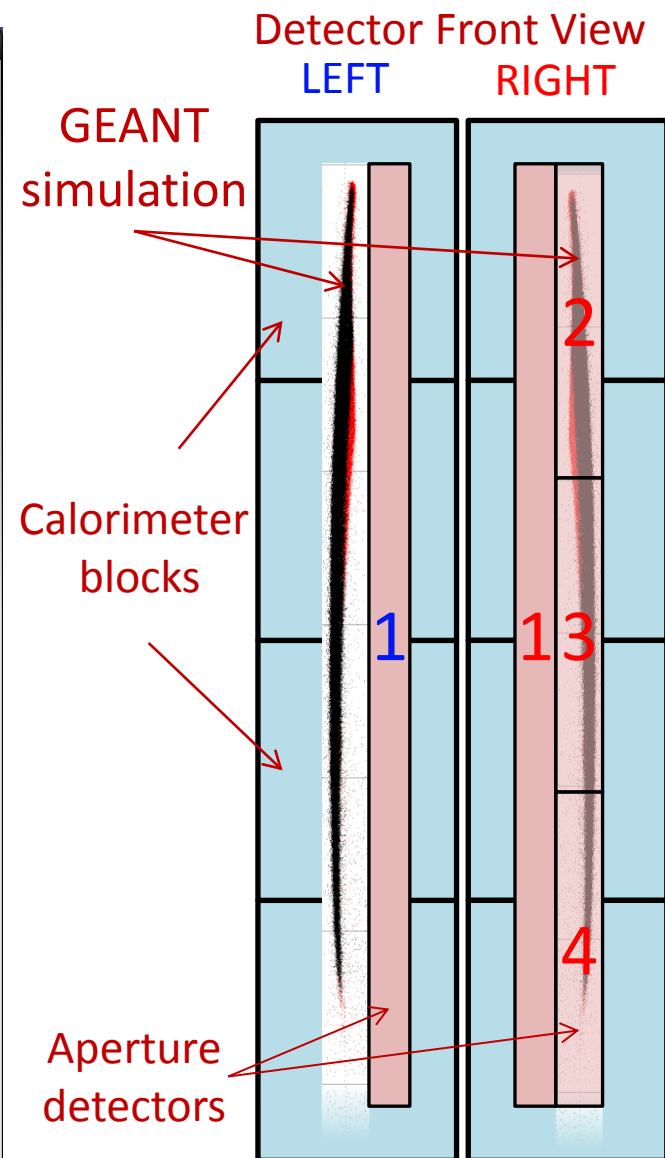
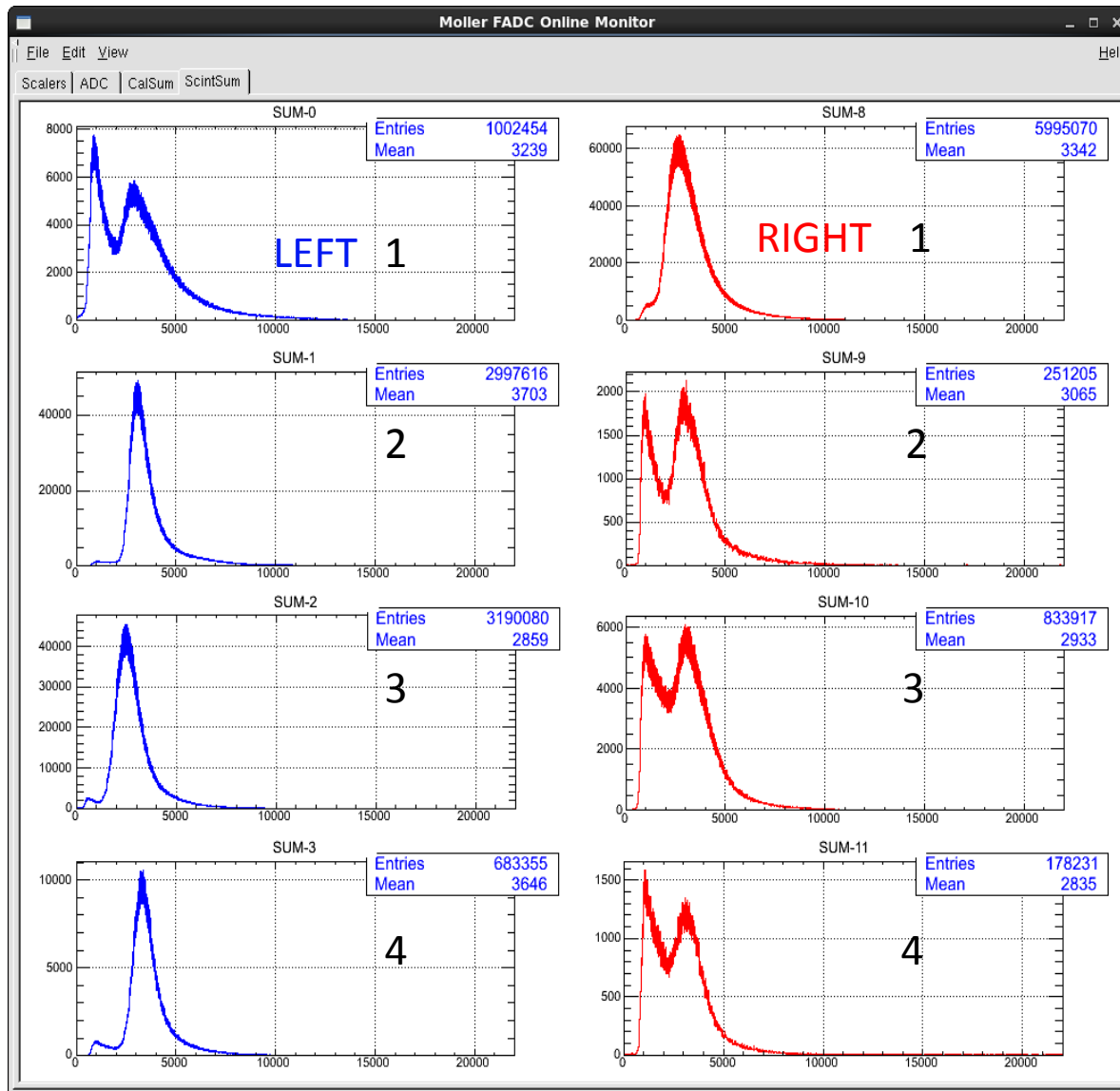
Beam quality was much better then in April commissioning run. But still not enough to do precise beam polarization measurements.



Target Position	6	5	4	3	2	1	0
Material	park	SM	Fe	Fe	SM	SM	Al
Thickness, μm		6.8	9.3	14.3	29.4	13.0	16.5

DETECTOR MISALIGNMENT? (April 02, 2014)

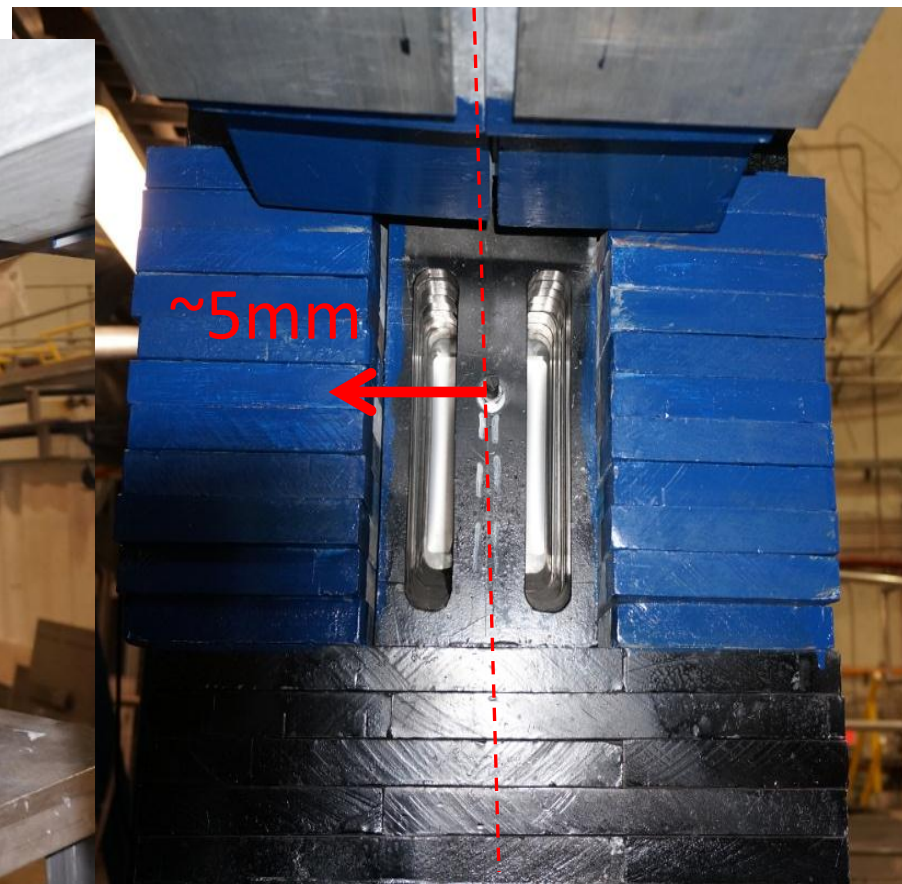
Amplitude spectrum from Aperture detectors (scintillators)



DETECTOR MISALIGNMENT? (cont.)

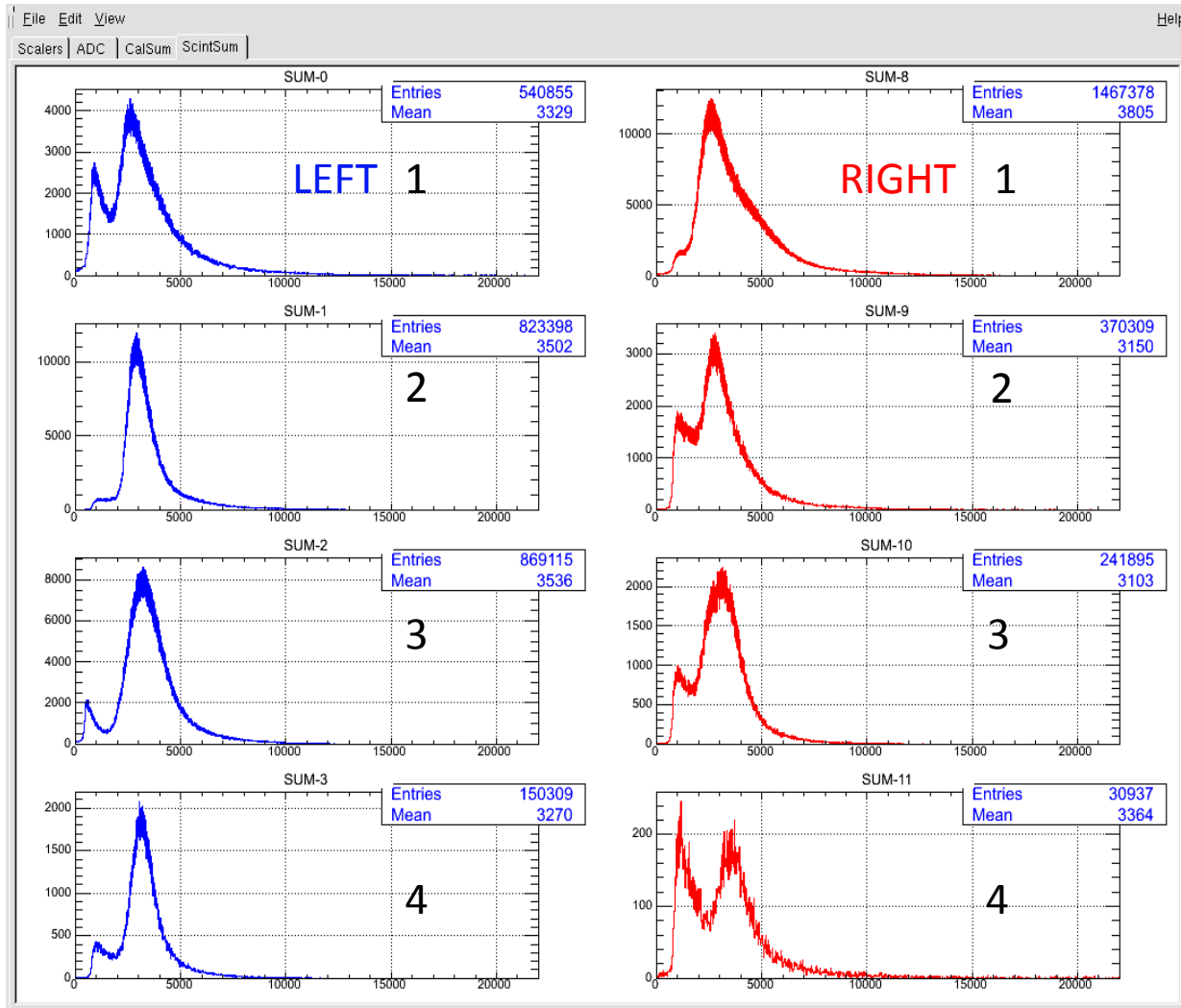
- Alignment group has found small offset of the detector: $\sim +2.5\text{mm}$.
- Offset of the dipole output collimator: $\sim -5\text{mm}$.
Total offset is $\sim 7.5\text{mm}$.
- Probably it is an explanation for this misalignment.
- Dipole output collimator offset was fixed

DIPOLE COLLIMATOR OFFSET



DETECTOR MISALIGNMENT? (December 08, 2014)

Amplitude spectrum from Aperture detectors (scintillators)



Looks better,
but still exists some
discrepancy with Geant
simulation between
number of events in
segment 1 and
segments 2, 3, 4
for right aperture detector.

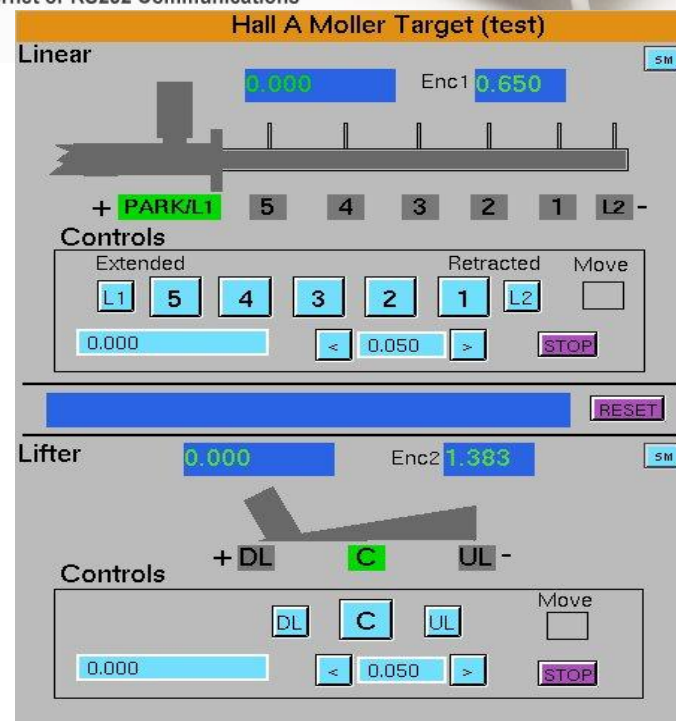
New controller for Moller target motion

- Old controller VME44 from Oregon Micro Systems is not supported with new version of EPICS
- New one is MAXnet controller with Ethernet port from the same company and with similar interface to step motors.
- William (Weiwei) developed EPICS software for new controller.
- Target motion was successfully tested with new EPICS software and new controller

MAXnet
Motion Controller



- 5-Axis
- 16-bit Analog Servo
- Backlash Compensation
- Designed for Expandability
- Ethernet or RS232 Communications



SUMMARY

1. Hall A Møller polarimeter is working.
2. Detector misalignment resolved?
3. Moller target motion with new EPICS software and controller has been tested.
4. Vadym is working on common ROOT based analyzer for old DAQ and FADC DAQ.
5. More systematic studies have to be done.

Thank you for your attention!

Amplitude spectrums from calorimeter (leadglass)

December 08, 2014

